COMP5152 Advanced Data Analytics

Hypothesis Testing and ANOVA

Raymond Pang
Department of Computing
The Hong Kong Polytechnic University

Recap

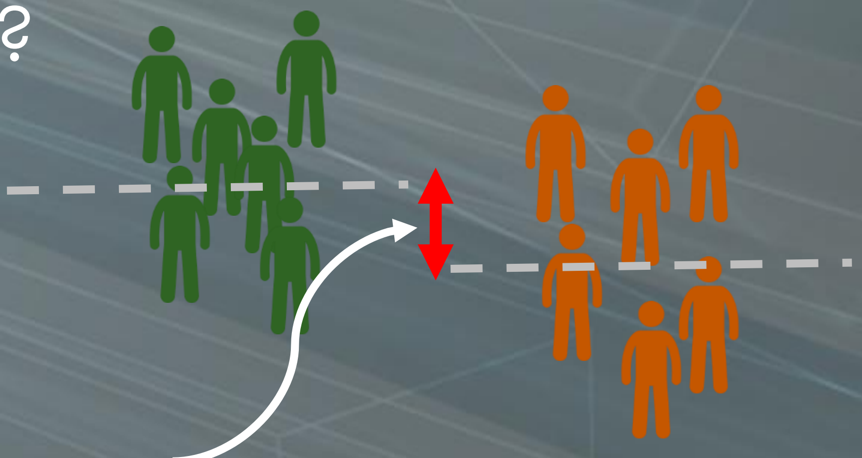
- In the last lecture, we understand what is statistics which tries to estimate a population based on smaller samples
- We learned basic statistics on how to measure or describe a group / sample
  - Mean, Mode, Medium
  - Variance and standard deviation

Hypothesis Testing

- Hypothesis testing is a fundamental statistical method for inferring population parameters based on sample data.
- It provides a structured approach for evaluating claims or assumptions about a population using empirical evidence.

The Motivation of Hypothesis Testing

- In a hypothesis testing, we are going to answer :
- **Is there a significant difference between 2 groups?**
- Or Just an Error?

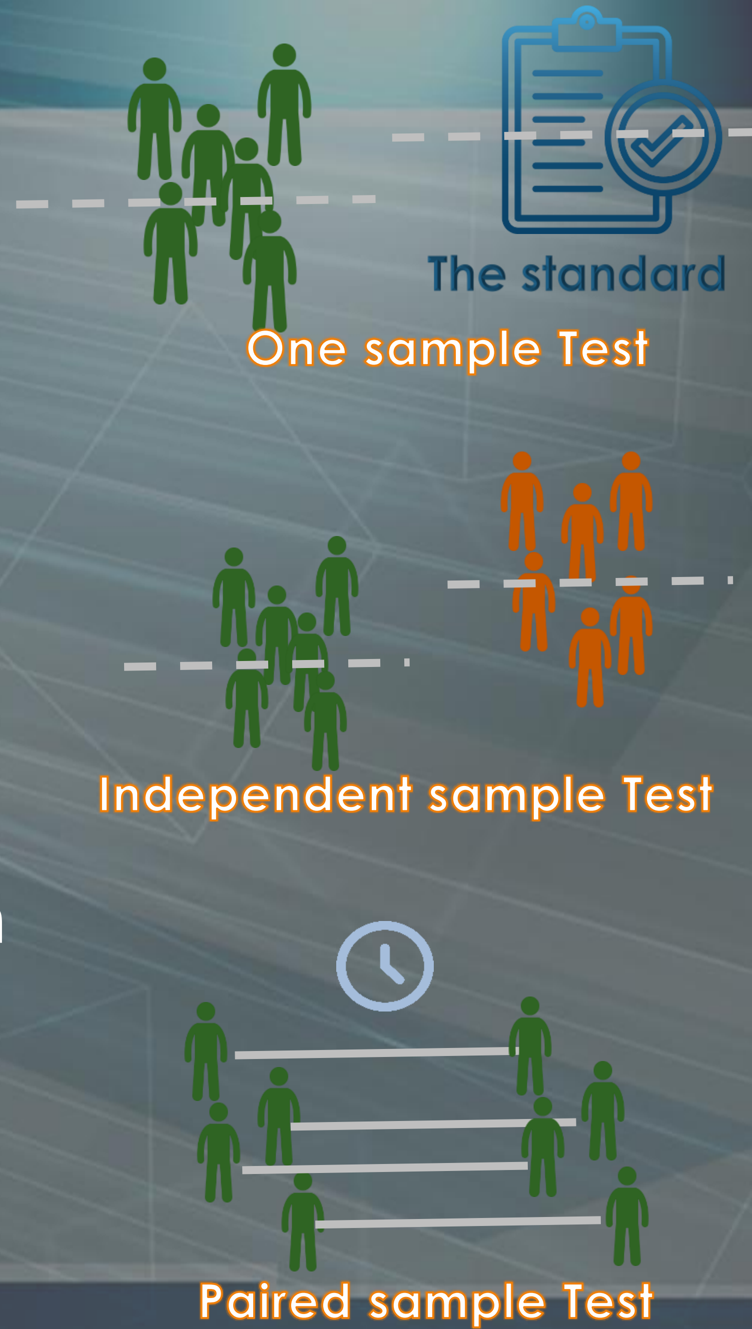


This is a significant difference?
Or
Simply a sampling error?

https://datatab.net/tutorial/t-test

Type of Tests

1. **One sample Test**
   - compare your data with a gold standard
2. **Independent sample Test**
   - compare two groups
3. **Paired sample Test**
   - compare two readings from same individuals (usually at two different times)



The t-Test

- t-test is a statistical test for comparing the means of two samples.
- Another well-known test is Z-test, which used for larger sample and known variance

Aspect	T-Test	Z-Test
Sample Size	Small (n < 30)	Large (n > 30)
Population Variance	Unknown	Known
Distribution Assumption (for the test statistic)	T-distribution - heavier tails	Normal distribution
Application	Small-sample studies, unknown variance	Large-sample studies, known variance

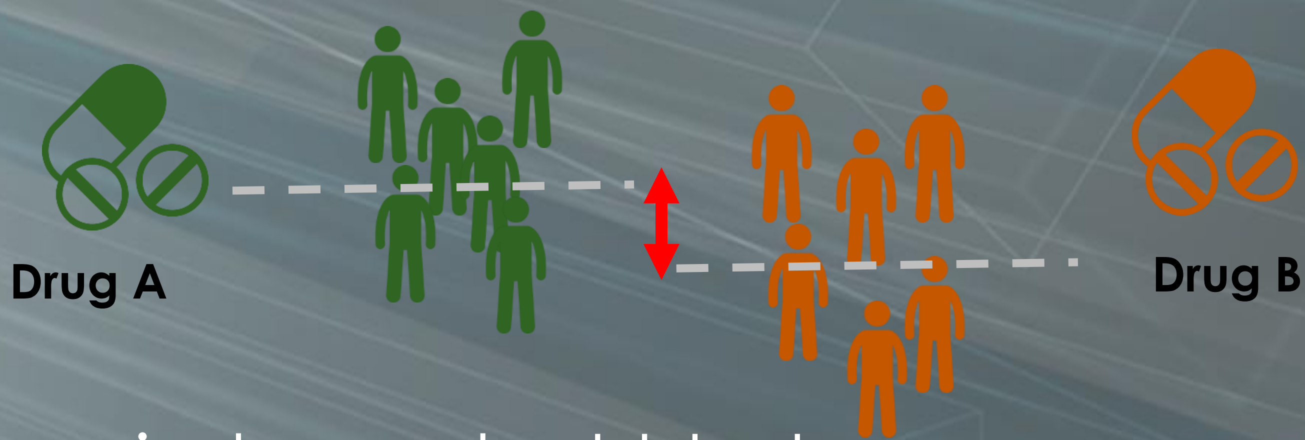
Example of a one sample t-test

A manufacturer of tissue paper claims that its tissue paper packet weighs 100 grams on average. To verify this, a sample of 30 packets is taken and weighed. The mean value of this sample is 97 grams.

We can perform a one sample t-test to see if the mean of 97 grams **is significantly different** from the claimed 100 grams.

Example of a t-test for independent samples

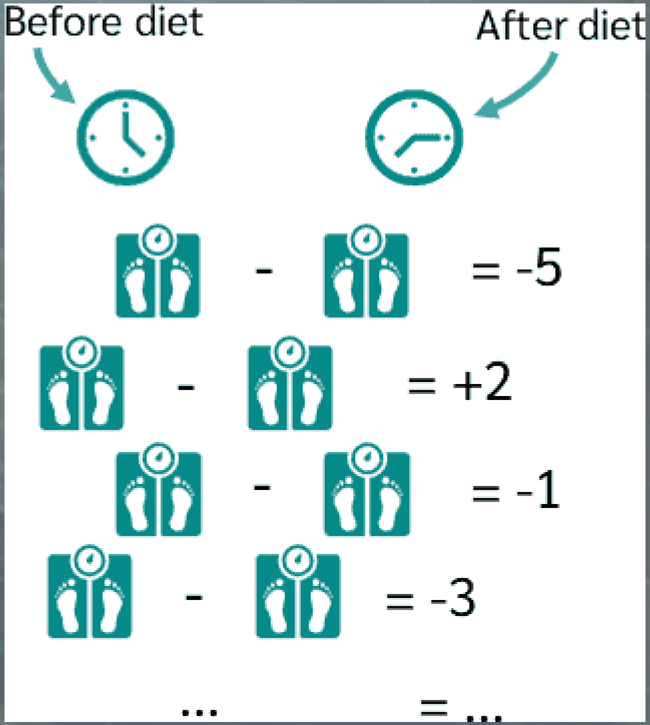
To compare the effectiveness of two drugs: A and B. We randomly divide 60 test subjects into two groups. The first group receives drug A, the second group receives drug B.



With an independent t-test we can now test whether there is **a significant difference** in pain relief between the two drugs.

Example of the t-test for paired samples

- We want to know how effective a diet is. To do this, we weigh 30 people before the diet and exactly the same people after the diet.
- Now we can see for each person how big the weight difference is between before and after. With a dependent t-test we can now check whether there is a significant difference.



Why and What is Hypothesis

- The generation of theories and testing them (confirmed or disconfirmed)
- Scientific statements are ones that can be **verified** and tested with reference to empirical evidence
- Non-scientific are statements are those that cannot be empirically tested.
- For example:

"cats are better than dogs" is non-scientific (no criteria to test against)

"cats are better than dogs at climbing trees" (now becomes testable)

Why and What is Hypothesis

- A good hypothesis is a **falsifiable** hypothesis! 可证伪假设

  e.g. "All horses are white"

- Remember "the absence of evidence is not evidence of absence".证据的缺失不代表证据不存在

If you have to prove that … – you look for ages, you see all horses are white in color, at the end of the search all you can say is that based on your observation **so far**, all horses are white. But there may be some horses hidden that are not white.

To falsify the hypothesis.. – is quite different. We can reject our hypothesis as soon as we see one horse in other color (e.g. black). We won't need to exhaustively check "ALL" horse, which is not feasible.

Why and What is Hypothesis

- Thus we have selected our null hypothesis (this says nothing is happening, nothing is different)
- You then need evidence to reject the null hypothesis and in favour of the alternative hypothesis
- Our work is easier and concrete when going to **reject** the Null hypothesis !!

Hypothesis for one sample test

- In the one sample t-test, the null hypothesis and the alternative hypothesis can be:

Null hypothesis H_0 : The sample mean is equal to the given reference value (so there is no difference).

Alternative hypothesis H_1 : The sample mean is not equal to the given reference value (so there is a difference).

Hypothesis for independent samples test

- In the independent t-test, hypotheses are:

Null hypothesis H_0: The means in the two groups are equal (so there is no difference between the two groups).

Alternative hypothesis H_1: The mean values in the two groups are not equal (i.e. there is a difference between the two groups).

Hypothesis for paired samples test

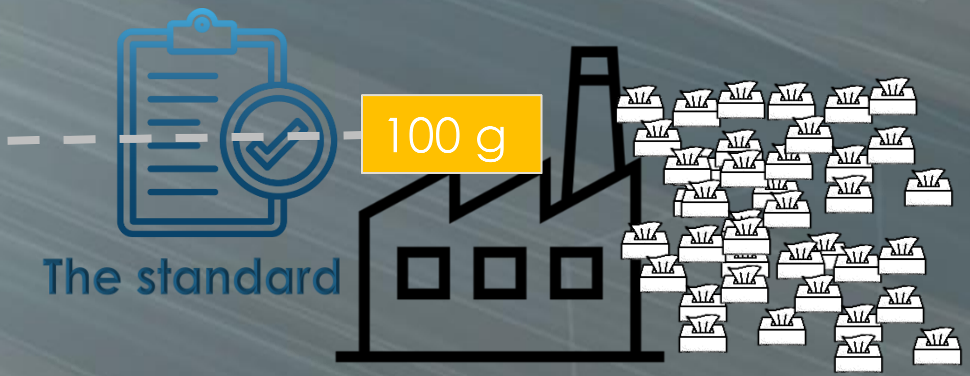
- In the paired t-test, the hypotheses are:

Null hypothesis H_0 : The mean of the differences between the pairs is zero.

Alternative hypothesis H_1 : The mean of the differences between the pairs is non-zero.

Formulating the One-sample t-Test

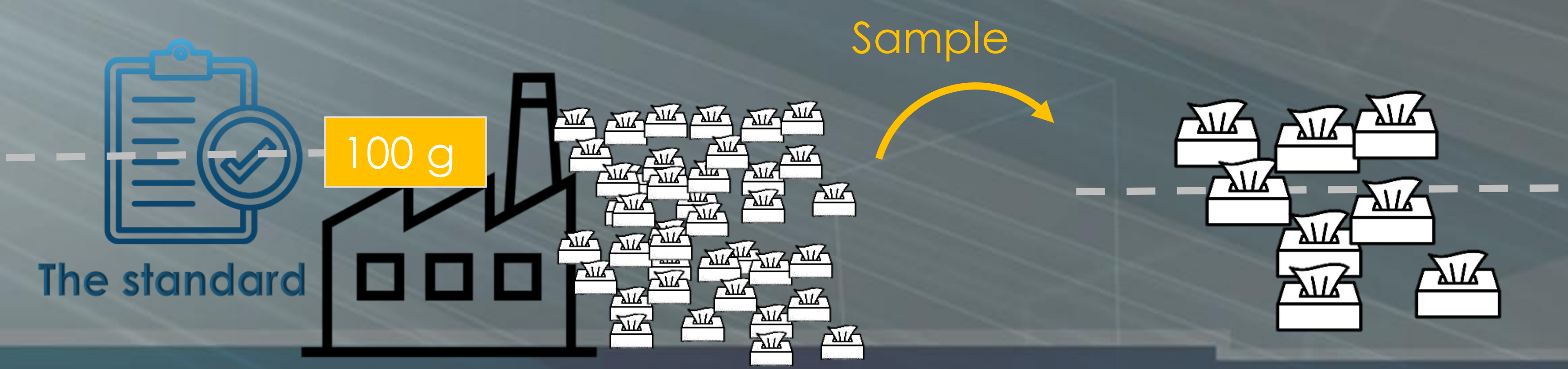
- Let's use again the tissue paper manufacturer example
- It claims that its tissue paper packet weighs 100 grams on average, which is actually the population mean (or the golden standard)



Formulating the One-sample t-Test

- Now, we sample 30 out of all packets
- With the t-test, we set our null hypothesis $\mathbf{H_0}$:

  there is **no difference** between sample mean to the population mean (the standard)

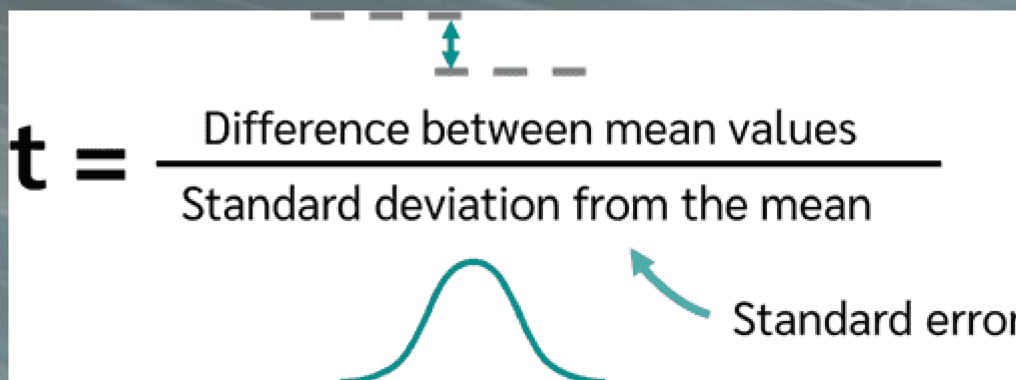


Formulating the One-sample t-Test

- You can imagine, every time we take a sample will have a variation
  - It would be very unlikely that we would draw a sample where the difference to the population mean is exactly zero.
- What t-test does is to measure at what difference level, we can say that there is a significant difference?
  - So we can reject the null hypothesis!!! Yeah!

Calculate the t-value

- To perform t-Test, we need to calculate a **t-value**:



Mean of the sample

Reference value

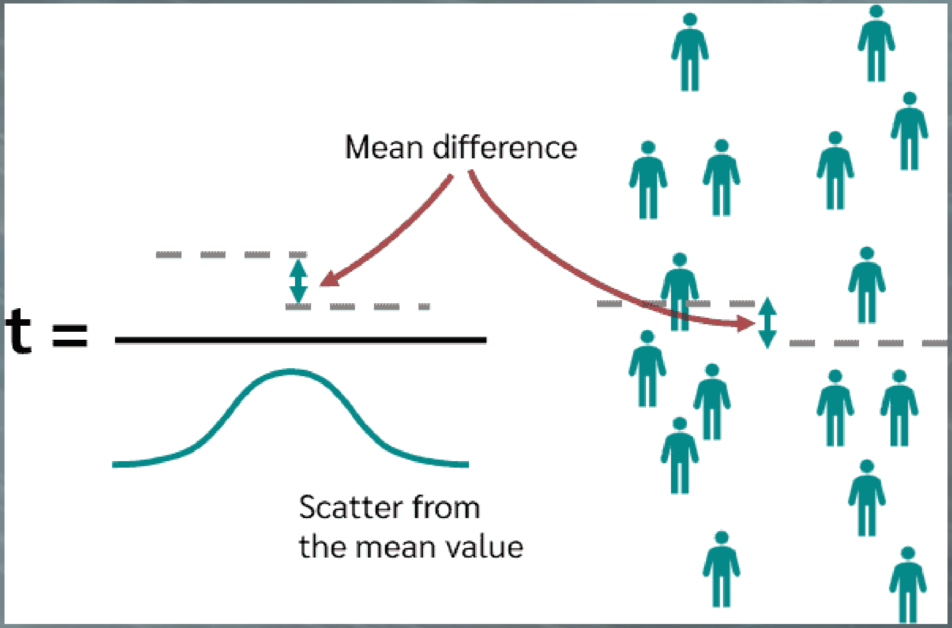
$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Standard deviation

Number of cases

Interpretation of t-value

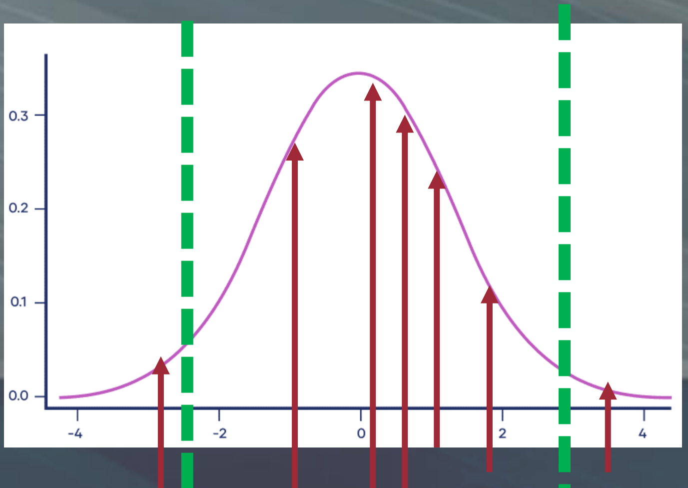
- t-value becomes larger the greater the difference between the means



- t-value becomes smaller if we have a larger dispersion of the mean values. So the greater the scatter of the data, the less a given mean difference matters!

Performing the One-sample t-Test

- Then, we compare the t-value with the **critical t-value**
- Remember every time we take a sample will have a variation
- So in the test, we need to set a level of significance
  - Can also understand as the cutoff we can tolerate (the variation)



Performing the One-sample t-Test

- This critical t-value can be read from the table (or use software to find it for you)
- Significance level
  - Usually 5%(0.05) is used
- Degrees of freedom

Critical values of *t* for two-tailed tests

Significance level (α)

Degrees of freedom *(df)*	.2	.15	.1	.05	.025	.01	.005	.001
1	3.078	4.165	6.314	12.706	25.452	63.657	127.321	636.619
2	1.886	2.282	2.920	4.303	6.205	9.925	14.089	31.599
3	1.638	1.924	2.353	3.182	4.177	5.841	7.453	12.924
4	1.533	1.778	2.132	2.776	3.495	4.604	5.598	8.610
5	1.476	1.699	2.015	2.571	3.163	4.032	4.773	6.869
6	1.440	1.650	1.943	2.447	2.969	3.707	4.317	5.959
7	1.415	1.617	1.895	2.365	2.841	3.499	4.029	5.408
8	1.397	1.592	1.860	2.306	2.752	3.355	3.833	5.041
9	1.383	1.574	1.833	2.262	2.685	3.250	3.690	4.781
10	1.372	1.559	1.812	2.228	2.634	3.169	3.581	4.587
11	1.363	1.548	1.796	2.201	2.593	3.106	3.497	4.437
12	1.356	1.538	1.782	2.179	2.560	3.055	3.428	4.318
13	1.350	1.530	1.771	2.160	2.533	3.012	3.372	4.221
14	1.345	1.523	1.761	2.145	2.510	2.977	3.326	4.140
15	1.341	1.517	1.753	2.131	2.490	2.947	3.286	4.073
16	1.337	1.512	1.746	2.120	2.473	2.921	3.252	4.015
17	1.333	1.508	1.740	2.110	2.458	2.898	3.222	3.965
18	1.330	1.504	1.734	2.101	2.445	2.878	3.197	3.922
19	1.328	1.500	1.729	2.093	2.433	2.861	3.174	3.883
20	1.325	1.497	1.725	2.086	2.423	2.845	3.153	3.850

Performing the One-sample t-Test

- In one sample t-test, degree of freedom df

  df = n - 1

  - where n is the number of samples
- So if we have a significance level of 5% and 9 degrees of freedom, we get a critical t-value of 2.262.
- If we calculate the t-value and obtain 2.5 which is larger than 2.262
  - thus the two means are so far apart that we can reject the null hypothesis $\mathbf{H_0}$.

The p-value

- Many literature do mention the p-value
- Also referred as probability value, tells you how likely it is that your data could have occurred under the null hypothesis.
- Thus, if p-value is small enough, we can reject the hypothesis
  - Usually $p<0.05$ (same as significance level)
- p-value can be calculated based on the t-value and degree of freedom.

Independent Sample and Paired Sample Tests

- The other 2 tests are more or less similar
- With different way to compute the t-value

t-test for independent samples

Mean sample 1

Mean sample 2

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Standard deviation Sample 1 and 2

Number of cases Sample 1 and 2

paired samples t-test

Mean of the difference

$$t = \frac{\overline{x_d} - 0}{\frac{s}{\sqrt{n}}}$$

Standard deviation

Number of cases

Example: Independent samples *t*-test

- Suppose we study the effect of caffeine on a motor test where the task is to keep the mouse centered on a moving dot.
- Everyone gets a drink; half get caffeine, half get placebo; nobody knows who got what.

Independent Sample Data

Experimental (Caff)	Control (No Caffeine)
12	21
14	18
10	14
8	20
16	11
5	19
3	8
9	12
11	13
	15
$N_1=9$, $M_1=9.778$, $SD_1=4.1164$	$N_2=10$, $M_2=15.1$, $SD_2=4.2805$

Independent Sample Test

1. Set significance level.

   Alpha = 0.05 (5%)

2. State Hypotheses.

   H_0: there is no effect of caffeine on a motor test, so the means in the both groups are equal

   $$\mu_1 = \mu_2.$$

   H_1:the means in the both groups are not equal

   $$\mu_1 \neq \mu_2.$$

Independent Sample Steps(2)

3. Calculate t-test statistic:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{SE_{diff}} = \frac{9.778 - 15.1}{1.93} = \frac{-5.322}{1.93} = -2.758$$

$$SE_{diff} = \sqrt{\frac{SD_1^2}{N_1} + \frac{SD_2^2}{N_2}} = \sqrt{\frac{(4.1164)^2}{9} + \frac{(4.2805)^2}{10}} = 1.93$$

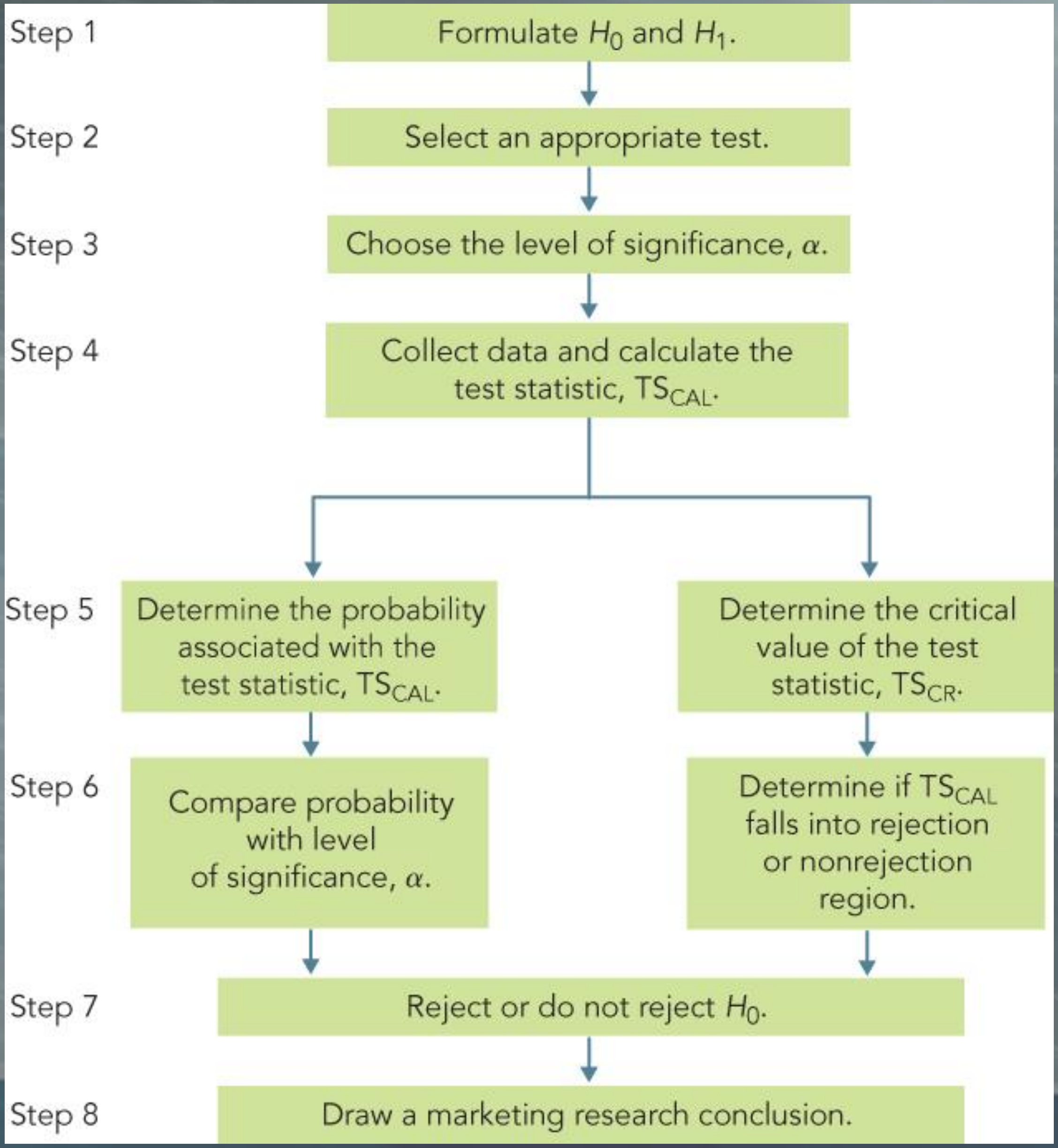
Independent Sample Steps (3)

4. Determine the critical value. Alpha is .05, 2 tails, and df = N1+N2-2 or 10+9-2 = 17. The value is 2.11.

5. State decision rule. If |-2.758| > 2.11, then reject the null.

6. Conclusion: Reject the null. the population means are different. Caffeine has an effect on the motor pursuit task.

Summary of Steps in Hypothesis Testing



Errors in Making Decisions

- Type I Error (**α**)
  - Rejects a true null hypothesis

  The probability of Type I Error is **α**
    - called level of significance,
    - set by researcher
- Type II Error (**β**)
  - Fails to reject a false null hypothesis
  - The probability of Type II Error is **β**
  - The power of the test is (1- β)
- The two probabilities are inversely related. Decreasing one increases the other, for a fixed sample size.

Result Probabilities

H_0: Innocent

H_1: Not Innocent

Jury Trial

	The Truth	
Verdict	Innocent	Guilty
Innocent	**Correct**	**Error**
Guilty	**Error**	**Correct**

Hypothesis Testing

	The Truth	
Decision	H_0 True	H_0 False
Do Not Reject H_0	**$1 - \alpha$**	**Type II Error (β)**
Reject H_0	**Type I Error (α)**	**Power $(1 - \beta)$**

How to Choose between Type I and Type II errors

- Choice depends on the significance or the cost of the errors
- Choose smaller Type I error when the cost of committing a Type I error is high
  - A criminal trial: convicting an innocent person

In practice, type II error is more complex to handle. So we would only focus on the manipulation of the Type I error, α .

Limitations of hypothesis test

- t-test or z-test
- Can only compare two groups/conditions
- Can only use one independent variable
- Anova (analysis of variance)
  - Developed by Fisher to overcome t-test limitations
  - Compare multiple groups at the same time

A/B Testing

A/B Testing

- A/B testing is primarily used in marketing, product development, and user experience research.
- Its purpose is to compare two versions (A and B) of a variable (e.g., a webpage, an ad, a product feature) to determine which one performs better in terms of a specified metric (e.g., click-through rate, conversion rate).

Why A/B Testing?

- Allows you to understand the causal impact of a product change
- Tests will provide data and empirical evidence to help you refine and enhance performance.
  - Design a more engaging customer experience (CX), write more compelling copy, and create more captivating visuals.
- As you continuously optimize, your marketing strategies will become more effective, increasing ROI and driving more revenue.

https://www.oracle.com/hk/cx/marketing/what-is-ab-testing/

Why A/B Testing

- A/B testing can be used to evaluate just about any digital marketing asset including:
  - Emails
  - Newsletters
  - advertisements
  - text messages
  - website pages
  - components on web pages
  - mobile apps

https://www.oracle.com/hk/cx/marketing/what-is-ab-testing/

A/B Testing: The Secret Behind the Leather Jacket (just 4 fun:)

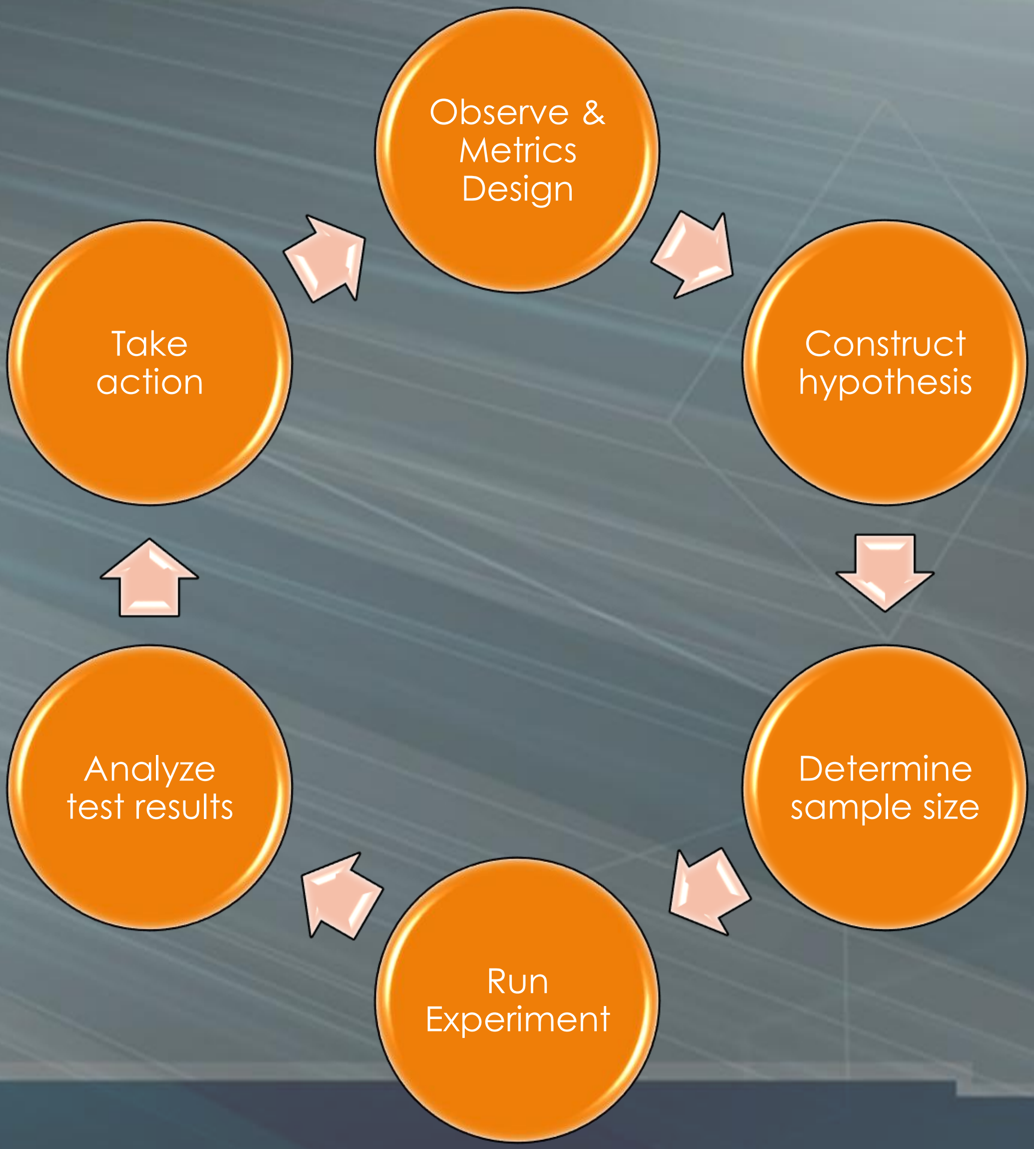
- Currently, NVIDIA occupies over 90% of the AI hardware market.
- It was also during this time that Huang began wearing leather jackets. Prior to this, his clothing style was very plain, even casual. After his daughter found a job at Louis Vuitton, she and his wife helped him with his image makeover.
- Unlike Steve Jobs' iconic turtleneck sweater, Jensen Huang, as an engineer, experimented with different clothing styles using an A/B testing approach. He constantly adjusted his image, just like optimizing a product.
- After four to five years of exploration, he found his ideal style and made his iconic appearance in a leather jacket when he launched AI in 2014. This engineer-like thinking is reflected not only in his product development but also in the shaping of his personal image.

From Intelligent Investor : "The author most trusted by Jensen Huang" deep discussion: the legend behind NVIDIA and the next step for AI.

When to use A/B Testing

- If you have a clear hypothesis
- If you can obtain sufficient sample size
- If you can test the hypothesis within a reasonable timeframe

Setup an A/B Test



Observe & Metrics Design

- Study and observe visitor behavior
  - E.g. visitor records, heatmaps
- Consider what can improve and designing an A/B test
- Choose the right metrics to measure based on your goals.
- For an e-commerce site, examples could include:
  - Conversion rate
  - Average order value
  - Subscriptions rate
  - Click-through-rate

Construct Hypothesis

- Clearly formulate null and alternative hypotheses that align with your goals.
- A/B test is an independent samples test
- E.g. Version A vs B: Enlarge the size of button will make it more prominent and will increase conversion
- H0 : Version A and B have the same conversion rate
- H1 :Version A and B has different conversion rate

Determine Sample Size

- How many subjects are needed for an A/B test?
- Affect how long the experiment has to run
- Usually has to be as large as thousands or more
  - Since you need at least a certain number of conversions (like hundreds)
- Use online A/B test sample size calculators
    - https://www.optimizely.com/sample-size-calculator/
  - Baseline conversion rate
    - current conversion rate for the product you are testing
  - Minimum detectable effect (MDE)
    - how much change from the baseline (how big or small a lift) you want to detect

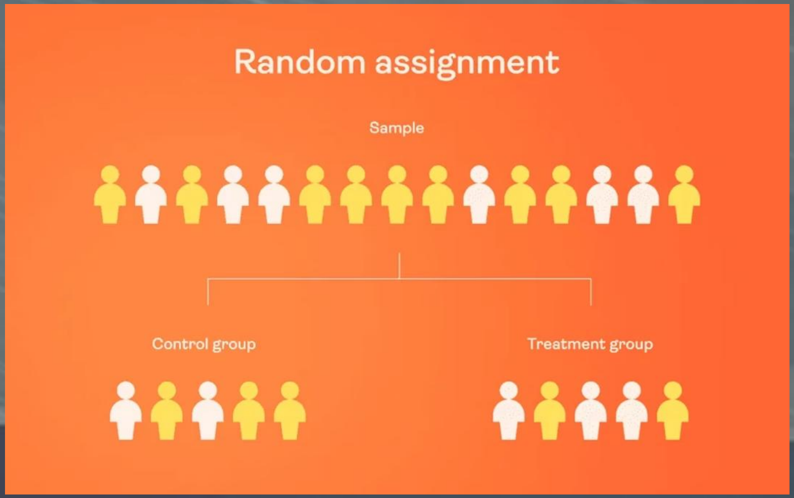
Determine Sample Size

- An example with a 20% baseline conversion rate and a 5% MDE.
- Based on these values, your experiment can detect 80% of the time when a variation's underlying conversion rate is actually 19% or 21% (20%, +/- 5% × 20%).
- If you try to detect differences smaller than 5%, your test is considered underpowered.

Run Experiment

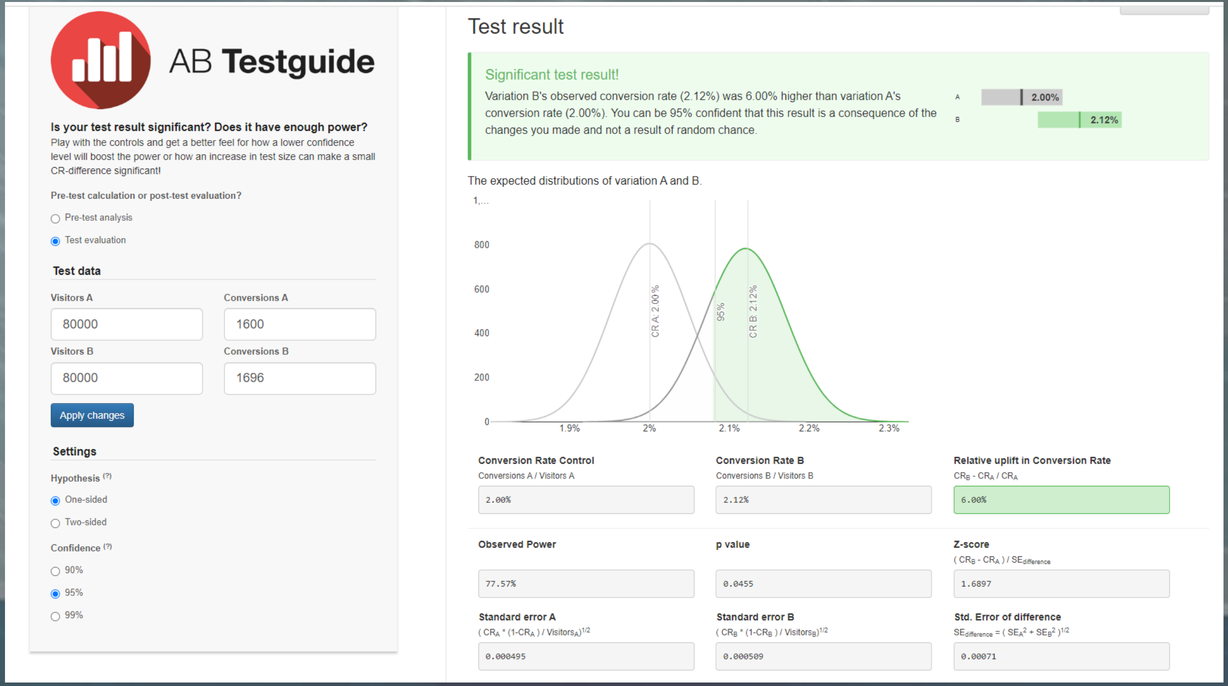
- Randomly divides user base into two groups:
  - 50% of users will see Version A.
  - 50% of users will see Version B.

- Test for a predetermined period, say 30 days. During this time, they collect data on user engagement metrics for both groups.



Analyze Test Results

- Does the conversion differ across two groups?
  - Paying attention to statistical significance!
- A simplest way is to rely on an A/B test calculator
  - E.g. AB Testguide: https://abtestguide.com/calc/



Analyze Test Results

- Inputs
  - Visitors and Conversions for Version A & B
  - One-side vs Two-side Hypothesis
  - Confidence
- Results
  - Significant level/ p-value (it will help you to reject H0)
  - Relative Uplift in Conversion
    - Your rate of improvements
  - Observed Power
    - the probability that the test correctly rejected the null hypothesis

Take Action

- Learn from your results
  - If there is a clear winner among the variations, go ahead with the implementation or changes.
  - If the test remains inconclusive, go back and rework on your hypothesis and prepare to start the next experiment

A/B Testing Consideration #1

- Be aware of statistical vs. practical significance
  - Statistical Significance – Observed mean differences are not likely due to sampling error
  - Practical Significance – Difference is large enough to actually matter

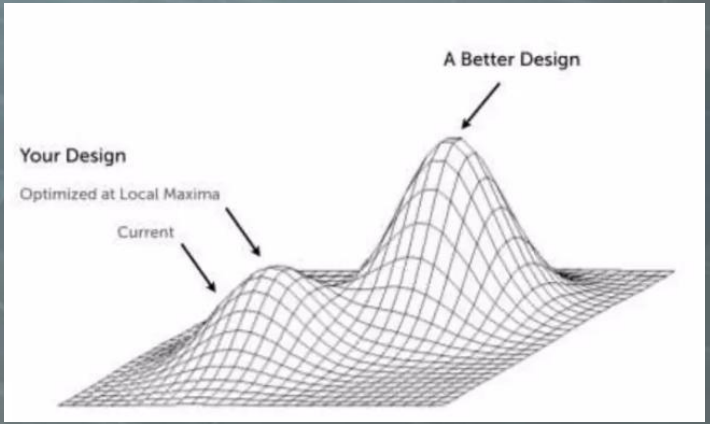
Experiment A → +0.1% Conversion Rate

A/B Testing Consideration #2

- Be aware of downstream metrics
  - Understand how your experiment impacts downstream metrics in unexpected ways
  - Example – Variant A increases upgrade rate by 10% (your target metric), but decreases LTV by 30% due to excessive monetization prompts

A/B Testing Consideration #3

- Be aware of the local maximum
  - Most A/B testing is done one variable at a time
  - If you continue to do this for a longer period of time, it will be impossible for you to arrive at a much better design



A/B Testing Consideration #4

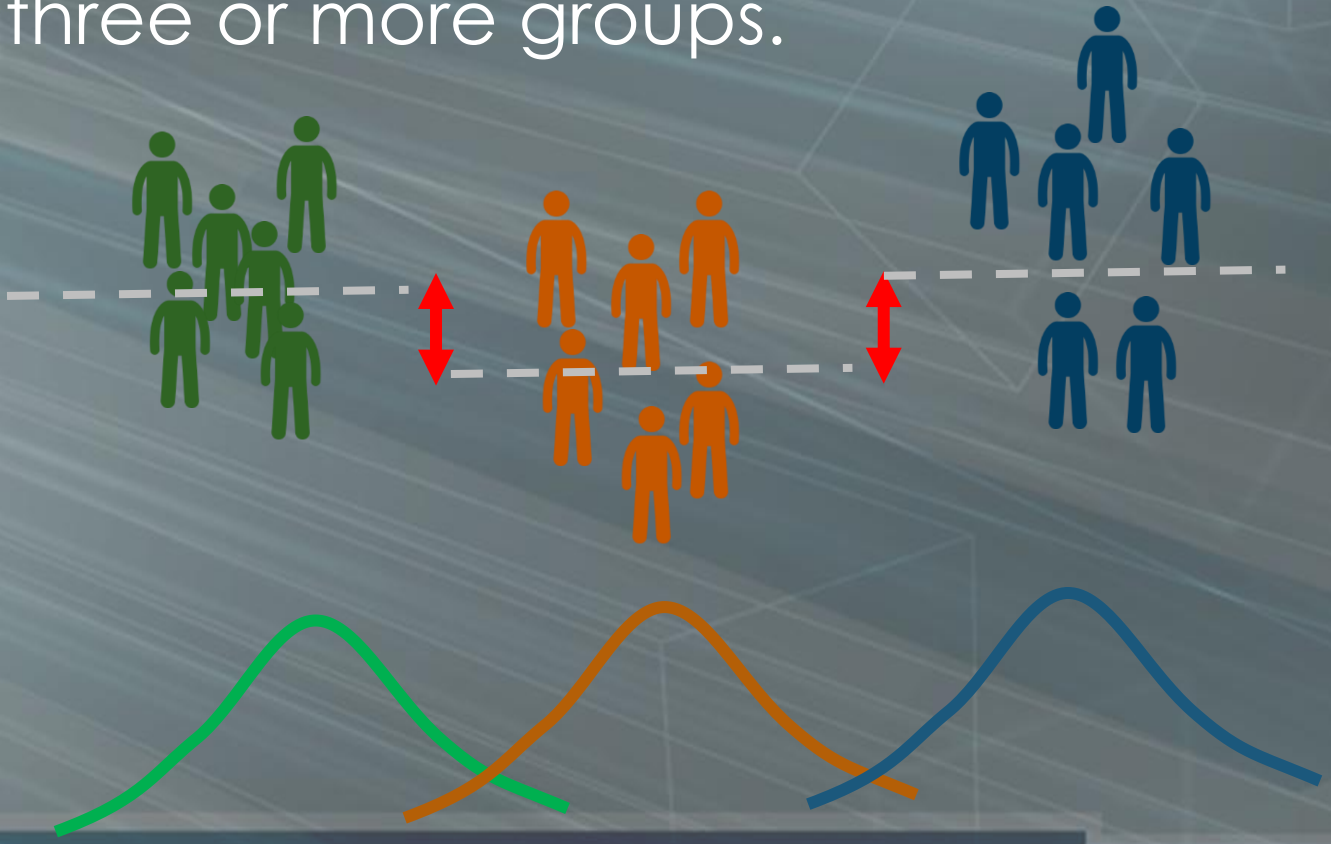
- Other common problems that can invalidate your experiment
  - Insufficient sample size
  - Insufficient time
  - Outlier events (i.e. Holidays)
  - Incorrect instrumentation
  - Bugs

ANOVA

https://www.datacamp.com/tutorial/anova-test

ANOVA

- ANOVA stands for Analysis of Variance, a statistical test used to compare the means of three or more groups.



Types of ANOVA Test

- One-way ANOVA (or One-factor)
  - there is one **independent variable** with two or more groups
  - E.g. effectiveness of the 3 different exercises is an independent variable

Independent variable

Dependent variable

Exercise

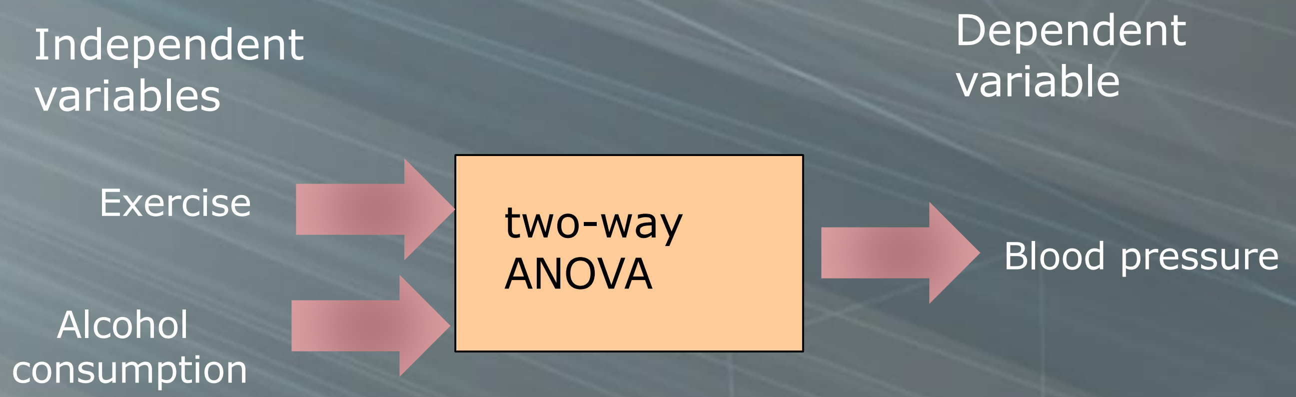
(Grouping variable)

One-way ANOVA

Blood pressure

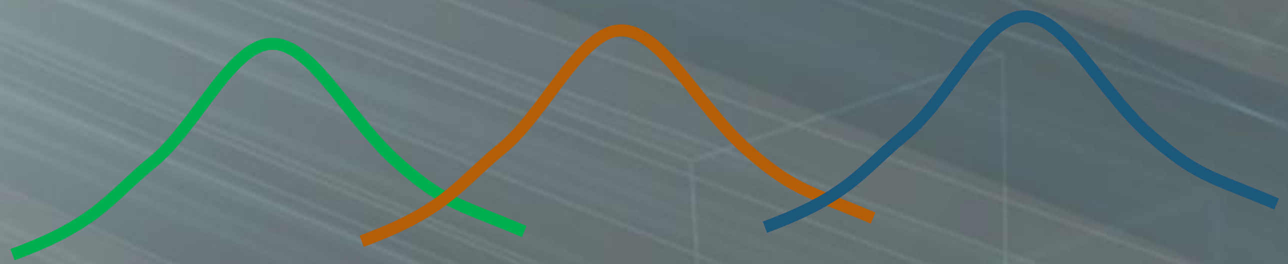
Types of ANOVA Test

- Two-way ANOVA (or Two factor)
  - there are two independent variables, each with two or more groups



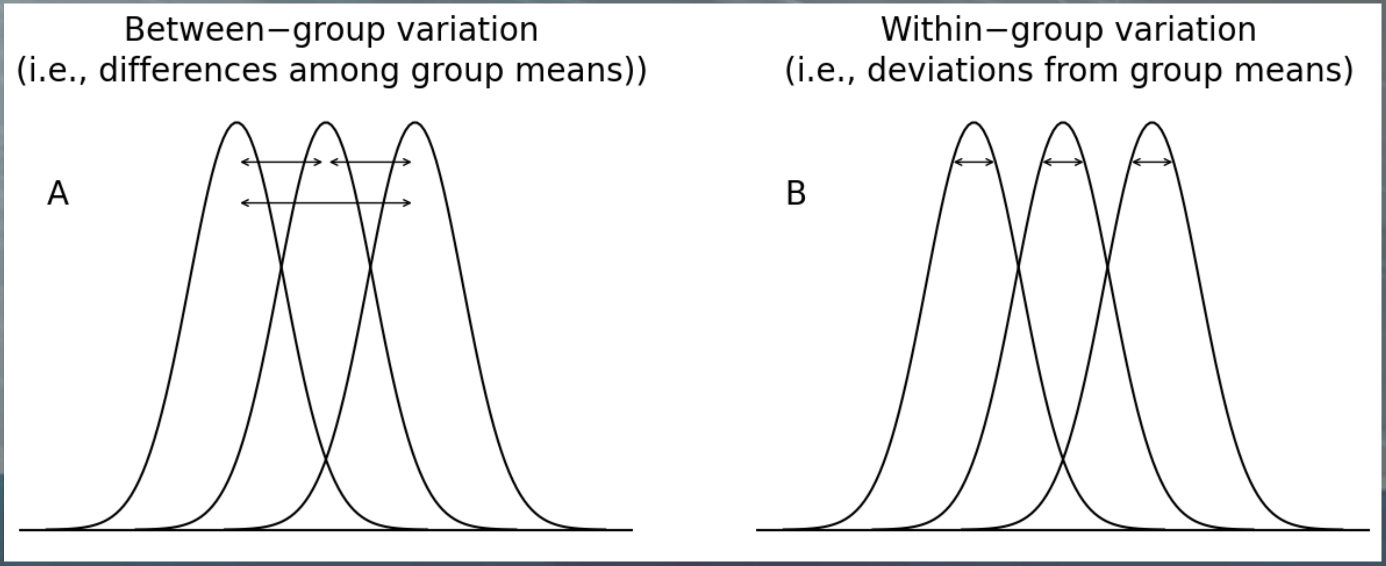
ANOVA

- Conceptually, ANOVA compares the variance **within** groups to the overall variance **between** all the groups to determine whether the groups appear distinct from each other or if they look quite the same.



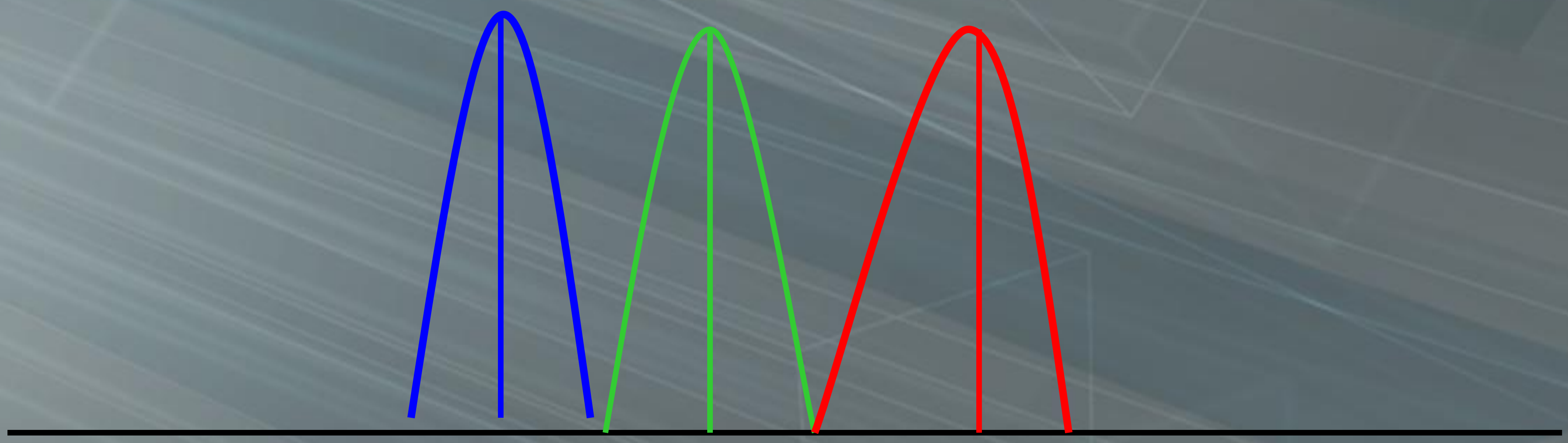
ANOVA

- **Within-Group Variability**: Variability caused by differences within individual groups, reflecting random fluctuations.
- **Between-Group Variability**: Variability caused by differences between the means of the different groups.



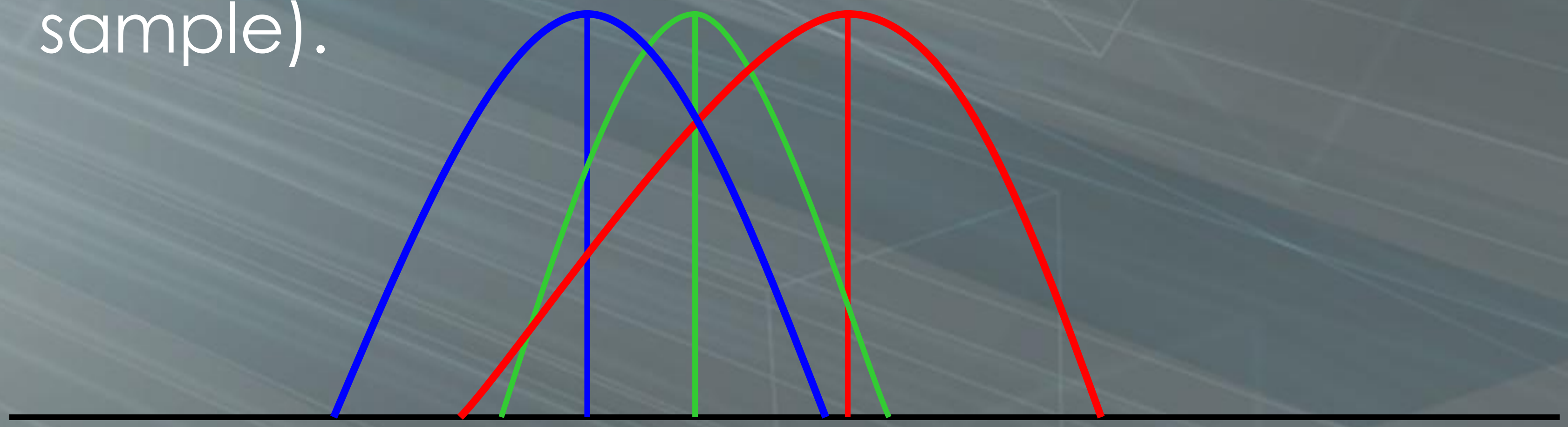
ANOVA

- When the groups have little variation within themselves, but large variation between them, it would appear that they are distinct and that their means are different.



ANOVA

- When the groups have a lot of variation within themselves, but little variation between them, it would appear that they are similar and that their means are not really different (perhaps they differ only because of peculiarities of the particular sample).



ANOVA

- F-statistics
  - F-statistic in one-way ANOVA is the ratio between the **mean square sum** between the groups and the mean square sum within the groups.
  - the larger the F-statistics, the greater the evidence that there is a difference between the group means.

$$F = \frac{Between\ group\ variance}{Within\ group\ variance}$$

$$F = \frac{MS_{between}}{MS_{within}}$$

Mean Squares Formulation

- Mean squares is the ratio of square sums to the degree of freedom.

$$MS_{between} = \frac{SS_{between}}{df_{between}} \qquad MS_{within} = \frac{SS_{within}}{df_{within}}$$

  - The degree of freedom between groups $df_{between}$ is equal to the number of groups minus one,
  - The degree of freedom within groups df_{within} is equal to the total number of participants minus the number of groups.

$$MS_{between} = \frac{SS_{between}}{K-1} \qquad MS_{within} = \frac{SS_{within}}{n-K}$$

n: number of observations
K: number of groups

Sum of Square Formulation

- Sum of squares within the group SS_{within} is the summation of the sum of squares for each group SS_{Group}

$$SS_{within} = \sum_{i=1}^{K} SS_{Group\ i}$$

n: number of observations
K: number of groups

group mean

$$SS_{Group\ i} = \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2$$

Sum of Square Formulation

- Sum of squares between the group $SS_{between}$ is the summation of the sum of squares (SS) for each group

$$SS_{between} = \sum_{i=1}^{K} n_i(\bar{x}_i - \bar{x})^2$$

Mean of the group i

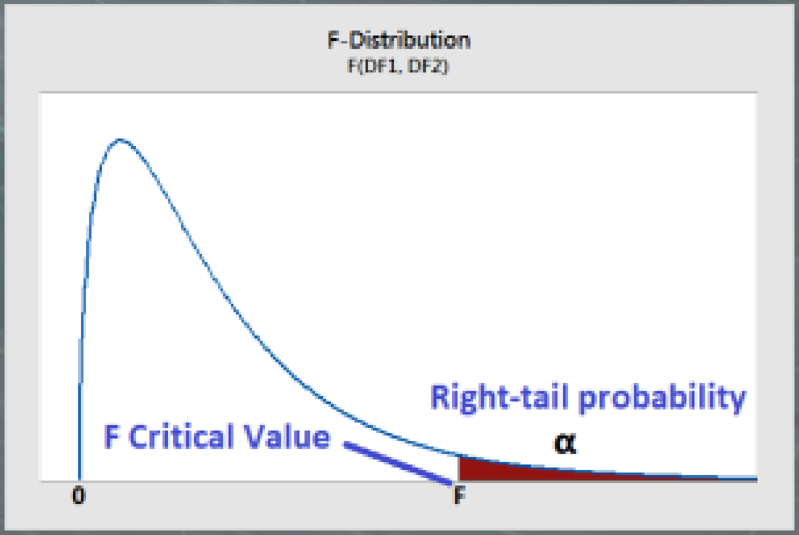
overall mean

n: number of observations
K: number of groups

F Critical Value

- Similar to t-test, f-test has also its critical value
- But, F-distributions require a numerator and denominator degrees of freedom (DF) to define its shape
- With also the significance level alpha, the F critical value is defined as

$$F_{K-1,n-K,\alpha}$$



F-table of Critical Values (alpha =0.05)

	F-table of Critical Values of α = 0.05 for F(df1, df2)																		
	DF1=1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**12**	**15**	**20**	**24**	**30**	**40**	**60**	**120**	**∞**
DF2=1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.95	248.01	249.05	250.10	251.14	252.20	253.25	254.31
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.37
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

Interpret the results

- **F-statistic:** The F-statistic measures the ratio of between-group variation to within-group variation.
- A higher F-statistic indicates a more significant difference between group means relative to random variation.

- **F critical value:** a threshold value that is used to determine whether the F statistic is statistically significant
- If F-statistics is larger than the F critical value, reject the null hypothesis and conclude that at least one group has a significantly different mean.

One-way ANOVA: Example

- Assume ”treatment results” from 13 patients visiting one of three doctors are given:

Doctor A	Doctor B	Doctor C
24	29	29
26	31	27
31	30	34
27	36	26
	33	

- H_0: The means are equal for all groups (The treatment results are from the same population of results)
- H_1: The means are different for at least two groups (They are from different populations)

1. Calculate the mean for each group and the overall mean

- Mean within groups:
  - Doctor A: 27
  - Doctor B: 31.8
  - Doctor C: 29
- Overall mean: $\dfrac{4 \cdot 27 + 5 \cdot 31.8 + 4 \cdot 29}{4 + 5 + 4} = 29.46$
- Variance around the mean matters for comparison.
- We must compare the variance within the groups to the variance between the group means.

2. Calculate the sum of squares for each group

- Sum of squares within groups:

$$SS_{Group\ i} = \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2$$

- Doctor A:

$$SS_{Group\ A} = (24 - 27)^2 + (26 - 27)^2 + \ldots = 26$$

- Doctor B:

$$SS_{Group\ B} = (29 - 31.8)^2 + (31 - 31.8)^2 + \ldots = 30.8$$

- Doctor C:

$$SS_{Group\ C} = (29 - 29)^2 + (27 - 29)^2 + \ldots = 38$$

3. Calculate the SS_{within}, $SS_{between}$

- Sum of squares within groups:

$$SS_{within} = \sum_{i=1}^{K} SS_{Group\ i} = 26 + 30.8 + 38 = 94.8$$

- Sum of squares between groups:

$$SS_{between} = \sum_{i=1}^{K} n_i(\bar{x}_i - \bar{x})^2$$

$$SS_{between} = 4(27 - 29.46)^2 + 5(31.8 - 29.46)^2 + 4(29 - 29.46)^2 = 52.43$$

4. Calculate the mean squares

- Obtain mean squares by dividing the number of degrees of freedom

$$MS_{within} = \frac{SS_{within}}{n-K} = \frac{94.8}{13-3} = 9.48$$

$$MS_{between} = \frac{SS_{between}}{K-1} = \frac{52.43}{3-1} = 26.2$$

5. Calculate the F-statistic and Interpret the results

- Finally, calculate f-statistics

$$F = \frac{MS_{between}}{MS_{within}} = \frac{26.2}{9.48} = 2.76$$

- Find the F critical value based on degree of freedoms and significance level (from the F-table)

$$F_{3-1,13-3,0.05} = 4.10$$

- $F < F_{critical}$, we can NOT reject the null hypothesis in our case.

ANOVA Table

Source of variation	Sum of squares	Deg. of freedom	Mean squares	F ratio
Between groups	$SS_{between}$	K-1	$MS_{between}$	$\frac{MS_{between}}{MS_{within}}$
Within groups	SS_{within}	n-K	MS_{within}	
Total	SS_{total}	n-1		

$$SS_{total}$$
$$= SS_{between} + SS_{within}$$
$$= (24 - 29.46)^2 + (26 - 29.46)^2 + \ldots + (26 - 29.46)^2$$

Assumptions in ANOVA

- The samples are independently drawn
- The population is normally distributed, with common variance
- They occur at random and independent of each other in the groups

Alternatives to ANOVA

- **Kruskal-Wallis Test:** It is a non-parametric alternative to one-way ANOVA when the normality assumption is violated.
- **MANOVA (Multivariate Analysis of Variance):** It extends the principles of ANOVA to multiple dependent variables. It tests whether the mean vectors of multiple dependent variables differ across groups.

Summary

- Hypothesis Testing
  - Three types of tests
  - Calculation of t-value and p-value in t-test
  - Type I and Type II errors
- A/B Testing
- ANOVA
  - Types of ANOVA
  - F-statistics